

■ Attributes

`Attributes[symbol]` gives the list of attributes for a symbol.

The attributes of a symbol can be set by assigning a value to `Attributes[s]`. If a single attribute is assigned, it need not be in a list. ■ `Attributes[s] = {}` clears all attributes of a symbol. ■ `Attributes[{s1, s2, ...}]` gives a list of the attributes for each of the *s_i*. ■ Attributes for functions must be set before any definitions that involve the functions are given. ■ The complete list of possible attributes for a symbol *f* is:

<code>Constant</code>	all derivatives of <i>f</i> are zero
<code>Flat</code>	associative
<code>HoldAll</code>	all the arguments of <i>f</i> are not evaluated
<code>HoldFirst</code>	the first argument of <i>f</i> is not evaluated
<code>HoldRest</code>	all but the first argument of <i>f</i> is not evaluated
<code>Listable</code>	<i>f</i> is automatically “threaded” over lists
<code>Locked</code>	attributes of <i>f</i> cannot be changed
<code>OneIdentity</code>	<i>f</i> [<i>a</i>], <i>f</i> [<i>f</i> [<i>a</i>]], etc. are equivalent to <i>a</i>
<code>Orderless</code>	commutative
<code>Protected</code>	values of <i>f</i> cannot be changed
<code>ReadProtected</code>	values of <i>f</i> cannot be read

■ See page 214. ■ See also: `SetAttributes`, `ClearAttributes`.